NIHAR HARDE

Data Science Undergraduate | Agentic AI, GenAI & Data Engineering
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PROFESSIONAL SUMMARY

Data Science undergraduate with hands-on experience building **agentic AI pipelines, GenAI/RAG systems, and scalable data solutions** using **Python, Java (OOP), and SQL**. Experienced across the complete software and data lifecycle—from architectural modularization to automated testing (**pytest**) and edge model optimization. Demonstrated track record of technical innovation in national hackathons (**Smart India Hackathon**) and peer-reviewed research published with **IEEE**.

EDUCATION

Shri Ramdeobaba College of Engineering and Management (RBU), Nagpur — Aug 2023 – May 2027
Bachelor of Technology, Computer Science and Engineering (Data Science) — CGPA: 8.33 / 10.00

The Royal Gondwana Public School — April 2023
Higher Secondary Education (Class XII) — Percentage: 83.4%

Bhavan's B.P. Vidya Mandir, Ashti — April 2021
Secondary Education, CBSE (Class X) — Percentage: 87.8%

RESEARCH PUBLICATION

Design and Implementation of a Legal Information Retrieval and Decision Support System Using RAG — Nov 2026
Uma Yadav, **Nihar Harde,** Piyush Patle, et al. | ***2026 24th OITS International Conference on Information Technology (OCIT), IEEE***

PROJECTS

Brief2Reel – Agentic AI Content Automation Pipeline — Jun 2026 – Present
Python, FastAPI, Multi-Agent Systems, ChromaDB, Groq SDK, Google Gemini API, PostgreSQL

- Owned the **Backend/Agents track** in a 4-person team, architecting an agentic workflow that translates structured product briefs into brand-grounded short video reels with human-in-the-loop review.
- Designed and implemented the **multi-agent orchestration** pipeline in **FastAPI**, sequencing Copywriter, Design, Audio, Video Compositor, and Critic/QA agents with structured schemas and automated fallback handling (Groq to Gemini).
- Engineered structural quality governance by deploying an **LLM-as-a-judge Critic Agent** with a 5-dimension scoring rubric and audit traces stored in **PostgreSQL** for end-to-end output traceability and brand consistency.
- Implemented first-principles guardrails, including a **semantic anti-repetition guard** using cosine distance (<0.85 threshold) and niche-partitioned **ChromaDB** collections for zero-hallucination context grounding.

Auto ATC – Automated Livestock Computer Vision & Scoring System — Sep 2025 – Dec 2025
Python, OpenCV, YOLOv8, LightGBM, ONNX Runtime, FastAPI, PyTorch | Smart India Hackathon

- Engineered an automated computer vision inference system for bovine breed classification and Body Condition Scoring (BCS), benchmarked across 41 domestic breeds against clinical standards.
- Developed under rapid **Smart India Hackathon sprint constraints**, applying structured problem-solving under pressure to convert complex veterinary criteria into practical automated scoring logic.
- Deployed a lightweight **FastAPI backend with ONNX Runtime INT8 quantization**, achieving low-latency CPU inference (<1.5s) and offline data synchronization for field deployment.

LexMind – Multi-lingual Information Retrieval & Decision Platform — Jan 2026 – Jun 2026
Python, FastAPI, ChromaDB, BM25, LangChain, Groq API, PostgreSQL, SQLAlchemy, Alembic, pytest, RAGAS

- Led a 3-person team developing an enterprise legal RAG system over Indian financial statutes (SEBI, RBI, PMLA), resulting in an accepted **IEEE OCIT 2026 research publication**.
- Engineered section-aware structural document chunking paired with hybrid dense-sparse retrieval (ChromaDB + BM25) to power context-aware query answering across trilingual queries (English, Hindi, Marathi).
- Architected a modular **FastAPI backend** with PostgreSQL persistence via SQLAlchemy/Alembic, input validation, and a 12-test automated **pytest verification suite**.
- Benchmarked pipeline retrieval reliability and answer accuracy systematically using the **RAGAS framework** with Azure OpenAI as an empirical evaluation judge.

Transaction-Level Data Pipeline & Analytics – FMCG Domain — Nov 2025
Databricks, PySpark, SQL, Delta Lake

- Architected the core end-to-end data pipeline on **Databricks** for transaction-level FMCG supply chain datasets, implementing **medallion architecture (Bronze to Silver to Gold)** to transform raw records into analytics-ready tables.
- Formulated complex **SQL queries** to uncover root causes behind delivery bottlenecks, and collaborated to design automated data quality validation and schema monitoring checks across pipeline stages.

TECHNICAL SKILLS

Languages & Core CS: Python, Java (OOP), C, SQL, Data Structures & Algorithms, Object-Oriented Design
Agentic AI & GenAI: Multi-Agent Systems, Agent Orchestration, Retrieval-Augmented Generation (RAG), LangChain, Prompt & Context Engineering, LLM-as-a-Judge Evaluation, RAGAS Benchmarking
Machine Learning & Computer Vision: scikit-learn, PyTorch, YOLOv8, OpenCV, LightGBM, ONNX Runtime, RFM Analytics, K-Means Clustering, Pandas, NumPy, Power BI
Data Engineering & Databases: PostgreSQL, SQLAlchemy, Alembic, MySQL, Databricks, PySpark, Delta Lake
Developer Tools & Backend: FastAPI, Git, GitHub, VS Code, Streamlit, Postman, Linux/CLI, Jupyter Notebook, MS Excel

RELEVANT COURSEWORK

Object-Oriented Programming (Java) | Data Structures & Algorithms | Database Management Systems | Cloud Computing (Azure & AWS) | Natural Language Processing | Machine Learning & Deep Learning | Data Warehousing & Business Intelligence

CERTIFICATES

Artificial Intelligence & Machine Learning with Drone Tech, TiHAN, IIT Hyderabad (Jun 2026) | Databases and SQL for Data Science with Python (Jan 2025) | Foundations of Data Science (Nov 2024) | Cloud Virtualization, Containers and APIs, Coursera (Nov 2025)